

Problem

A positive sequence wye-connected source where $V_{an} = 120 \angle 90^\circ \text{ V}$, is connected to a delta-connected load where $Z_L = (60 + j45) \Omega$. Determine the line currents.

Step-by-step solution

Step 1 of 5

Convert the delta connected load to wye connection.

Calculate the impedance of the wye connected load.

$$Z_Y = \frac{Z_\Delta}{3}$$

Substitute $60 + j45 \Omega$ for Z_Δ .

$$\begin{aligned} Z_Y &= \frac{60 + j45}{3} \\ &= 20 + j15 \Omega \end{aligned}$$

Therefore, the impedance of the wye connected load is $20 + j15 \Omega$.

Step 2 of 5

Consider the circuit diagram shown in Figure 1.

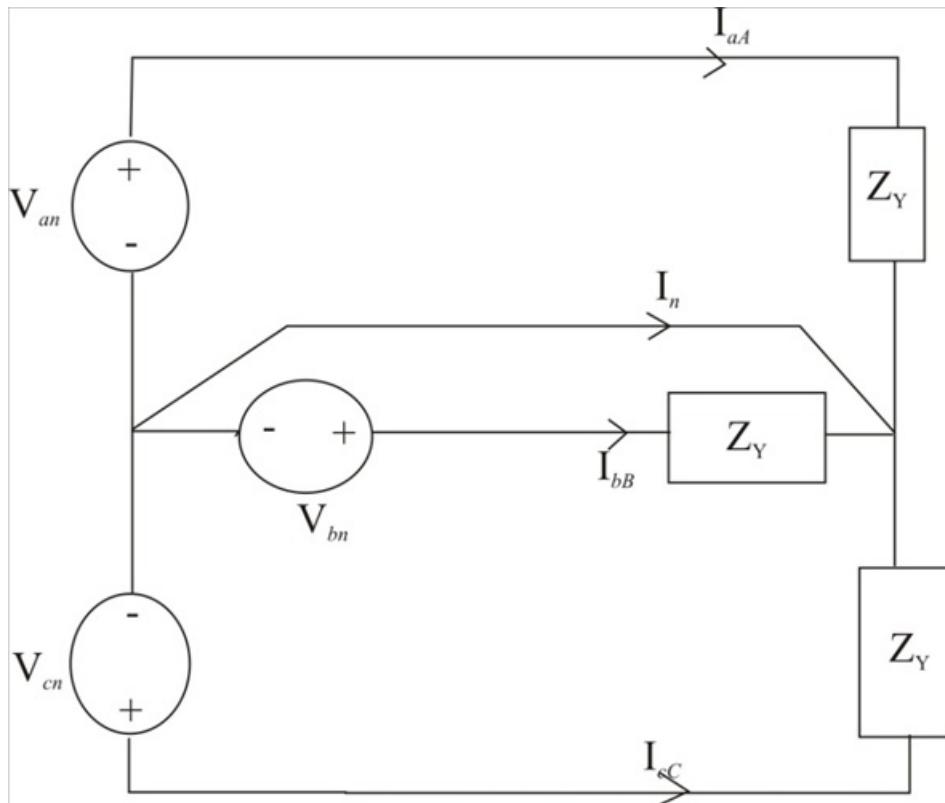


Figure 1

Step 3 of 5

Calculate the value of the line current I_{aA} .

$$I_{aA} = \frac{V_{an}}{Z_Y}$$

Substitute $20 + j15 \Omega$ for Z_Y and $120\angle 90^\circ \text{ V}$ for V_{an} .

$$\begin{aligned} I_{aA} &= \frac{120\angle 90^\circ}{20 + j15} \\ &= \frac{120\angle 90^\circ}{25\angle 36.87^\circ} \\ &= 4.8\angle 53.13^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_{aA} is $4.8\angle 53.13^\circ$.

Compare with $I_L\angle\theta^\circ$.

The values of I_L and θ° are 4.8 A and 53.13° respectively.

Step 4 of 5

In a positive sequence the current I_{aA} leads I_{bB} which in turn leads I_{cC} .

Calculate the value of the current I_{bB} .

$$I_{bB} = I_L\angle\theta^\circ - 120^\circ$$

Substitute 53.13° for θ° and 4.8 A for I_L .

$$\begin{aligned} I_{bB} &= 4.8\angle 53.13^\circ - 120^\circ \\ &= 4.8\angle -66.87^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_{bB} is $4.8\angle -66.87^\circ \text{ A}$.

Step 5 of 5

Calculate the value of the current I_{cC} .

$$I_{cC} = I_L\angle\theta^\circ + 120^\circ$$

Substitute 53.13° for θ° and 4.8 A for I_L .

$$\begin{aligned} I_{cC} &= 4.8\angle 53.13^\circ + 120^\circ \\ &= 4.8\angle 173.13^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_{cC} is $4.8\angle 173.13^\circ \text{ A}$.